Institut Sabadell

Actividad 1: Instalación de Servidor de Correo y Webmail

CFGS Administració de sistemes informàtics i xarxes

Mòdul 08: Serveis de xarxa

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Introducción

En esta actividad se implementa un **sistema completo de correo electrónico en red local**, compuesto por:

- **Servidor SMTP** → Envío de correos (Postfix)
- **Servidor POP3/IMAP** → Recepción de correos (Dovecot)
- **Servidor DNS** → Resolución de nombres (Bind9)
- **Webmail** → Acceso vía navegador (Roundcube + Apache + MySQL)

Componentes

- **Postfix:** Servidor de correo que gestiona el envío de emails mediante el protocolo SMTP.
- **Dovecot:** Servidor que permite a los usuarios recibir correos mediante POP3 o IMAP.
- **Roundcube:** Cliente web que permite consultar el correo desde un navegador.
- **Apache HTTP Server:** Servidor web que aloja la interfaz de Roundcube.
- **MySQL:** Sistema gestor de bases de datos utilizado por Roundcube.
- **BIND:** Servidor DNS encargado de resolver nombres de dominio en la red local.

Objetivos de la Práctica

- Configurar correctamente el hostname del servidor
- Instalar y configurar un servidor SMTP (Postfix)
- Crear cuentas locales de correo
- Configurar acceso POP3 e IMAP
- Configurar resolución DNS local
- Instalar y configurar un sistema Webmail
- Verificar el funcionamiento completo del sistema

Desarrollo de la Práctica

1. Comprobar y Configurar el Hostname

Verificar hostname actual

- hostname

Cambiar el hostname

- sudo hostnamectl set-hostname {dominio_principal}
- sudo hostnamectl set-hostname --static {dominio_principal}

Confirmar el cambio

- hostname

NOTA: hostname debe coincidir con el dominio configurado en el servidor de correo y DNS.

2. Instalar y Configurar Postfix (Servidor SMTP)

Instalación

- sudo apt-get install postfix

```
user@viejofraile:~$ sudo apt-get install postfix
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libnsl2
Suggested packages:
  postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver
  postfix-mysql postfix-pcre postfix-pgsql postfix-sqlite procmail sasl2-bin
  | dovecot-common
The following NEW packages will be installed:
  libnsl2 postfix
0 upgraded, 2 newly installed, 0 to remove and 268 not upgraded.
Need to get 1.296 kB of archives.
After this operation, 4.321 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libnsl2 amd64 1.3.0-3bui
ld3 [41,4 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 postfix amd64 3.8.6-1bui
ld2 [1.254 kB]
```

Durante la instalación:

- Tipo: **Internet Site**
- Nombre del sistema de correo: {dominio_principal}

Comprobar dominio configurado

- cat /etc/mailname

```
user@viejofraile:~$ cat /etc/mailname
viejofraile.local
```

Si se modifica:

- sudo systemctl restart postfix

```
user@viejofraile:~$ sudo systemctl restart postfix
user@viejofraile:~$
```

3. Añadir Usuarios de Correo

- sudo adduser bob
- sudo adduser alice

Estos usuarios podrán enviar y recibir correo local.

```
user@viejofraile: ~
File  Edit  View  Search  Terminal  Help
user@viejofraile:~$ sudo adduser user_1
info: Adding user `user_1' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user_1' (1001) ...
info: Adding new user `user_1' (1001) with group `user_1 (1001)' ...
info: Creating home directory `/home/user_1' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user_1
Enter the new value, or press ENTER for the default
        Full Name []: USUARIO 1
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
info: Adding new user `user_1' to supplemental / extra groups `users' ...
info: Adding user `user_1' to group `users' ...
user@viejofraile:~$
```

```
File  Edit  View  Search  Terminal  Help
user@viejofraile:~$ sudo adduser user_2
[sudo] password for user:
info: Adding user `user_2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user_2' (1002) ...
info: Adding new user `user_2' (1002) with group `user_2 (1002)' ...
info: Creating home directory `/home/user_2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user_2
Enter the new value, or press ENTER for the default
        Full Name []: USUARIO 2
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n]
info: Adding new user `user_2' to supplemental / extra groups `users' ...
info: Adding user `user_2' to group `users' ...
user@viejofraile:~$
```

4. Instalar Mailx (Cliente de correo en consola)

- sudo apt-get install bsd-mailx

```
user@viejofraile:~$ sudo apt-get install bsd-mailx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  liblockfile-bin liblockfile1
The following NEW packages will be installed:
  bsd-mailx liblockfile-bin liblockfile1
0 upgraded, 3 newly installed, 0 to remove and 268 not upgraded.
Need to get 85,2 kB of archives.
After this operation, 251 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Permite enviar correos desde terminal para pruebas.

5. Comprobar Logs del Servidor

- cat /var/log/mail.log

```
user@viejofraile:~$ cat /var/log/mail.log
2026-02-26T11:25:53.025608+01:00 user-VirtualBox postfix/postfix-script[5507]: starting the Postfix mail system
2026-02-26T11:25:53.025876+01:00 user-VirtualBox postfix/master[5509]: daemon started -- version 3.8.6, configuration /etc/postfix
2026-02-26T11:26:26.402174+01:00 viejofraile postfix/postfix-script[5601]: stopping the Postfix mail system
2026-02-26T11:26:26.408719+01:00 viejofraile postfix/master[5509]: terminating on signal 15
2026-02-26T11:26:28.105798+01:00 viejofraile postfix/postfix-script[5934]: starting the Postfix mail system
2026-02-26T11:26:28.117488+01:00 viejofraile postfix/master[5936]: daemon started -- version 3.8.6, configuration /etc/postfix
```

Aquí se verifican errores o envíos correctos.

6. Instalar Dovecot (POP3)

- sudo apt-get install dovecot-pop3d

```
user@viejofraile:~$ sudo apt-get install dovecot-pop3d
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  dovecot-core
Suggested packages:
  dovecot-gssapi dovecot-imapd dovecot-ldap dovecot-lmtpd dovecot-managesieved
  dovecot-mysql dovecot-pgsql dovecot-sieve dovecot-solr dovecot-sqlite
  dovecot-submissiond ntp
The following NEW packages will be installed:
  dovecot-core dovecot-pop3d
0 upgraded, 2 newly installed, 0 to remove and 268 not upgraded.
Need to get 3.388 kB of archives.
After this operation, 10,8 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dovecot-core amd
64 1:2.3.21+dfsg1-2ubuntu6.1 [3.351 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dovecot-pop3d am
d64 1:2.3.21+dfsg1-2ubuntu6.1 [37,2 kB]
```

7. Configurar POP3 en Red Local

Configurar Postfix

Editar:

- sudo nano /etc/postfix/main.cf

Agregar o modificar:

mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128 [Añadir dirección de red local]
home_mailbox = Maildir/
mailbox_command =

Reiniciar:

- sudo systemctl restart postfix

Configurar Dovecot

Editar:

- sudo nano /etc/dovecot/conf.d/10-auth.conf

```
  GNU nano 7.2              /etc/dovecot/conf.d/10-auth.conf
##
## Authentication processes
##

# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGINDISABLED capability). Note that if the remote I
# matches the local IP (ie. you're connecting from the same computer),
# connection is considered secure and plaintext authentication is allow
# See also ssl=required setting.
disable_plaintext_auth = no
```

Modificar:

- disable_plaintext_auth = no

Editar:

- sudo nano /etc/dovecot/conf.d/10-mail.conf

Modificar:

- mail_location = maildir:~/Maildir

```
  GNU nano 7.2              /etc/dovecot/conf.d/10-mail.conf
#   %n - user part in user@domain, same as %u if there's no domain
#   %d - domain part in user@domain, empty if there's no domain
#   %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
#   mail_location = maildir:~/Maildir
#   mail_location = mbox:~/mail:INBOX=/var/mail/%u
#   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
#
# <doc/wiki/MailLocation.txt>
#
mail_location = mbox:~/mail:INBOX=/var/mail/%u
```

Reiniciar:

- sudo systemctl restart dovecot

- sudo systemctl status dovecot

8. Configuración DNS Local

Editar zona:

- sudo nano /var/cache/bind/db.{tudominio.local}

Agregar:

```
{tudominio.local} IN MX 10 mail.{tudominio.local}.
pop3    IN CNAME server
smtp    IN CNAME server
```

Reiniciar DNS:

- sudo systemctl restart bind9

Comprobar:

- dig -t MX {tudominio.local}

-

Limpiar caché:

- sudo resolvectl flush-caches

9. Instalación IMAP y Webmail

Instalar MySQL

sudo apt-get install mysql-server

```
root@viejofraile:/home/user# sudo apt-get install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libevent-core-2.1-7t64 libevent-pthreads-2.1-7t64 libhtml-template-perl
  libmecab2 libprotobuf-lite32t64 libprotobuf32t64 mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0
  mysql-common mysql-server-8.0 mysql-server-core-8.0
Suggested packages:
  libipc-sharedcache-perl tinyca
The following NEW packages will be installed:
  libevent-core-2.1-7t64 libevent-pthreads-2.1-7t64 libhtml-template-perl
  libmecab2 libprotobuf-lite32t64 mecab-ipadic mecab-ipadic-utf8 mecab-utils
  mysql-client-8.0 mysql-client-core-8.0 mysql-common mysql-server
  mysql-server-8.0 mysql-server-core-8.0
The following packages will be upgraded:
  libprotobuf32t64
1 upgraded, 14 newly installed, 0 to remove and 267 not upgraded.
Need to get 30,0 MB of archives.
After this operation, 241 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Instalar IMAP

sudo apt-get install dovecot-imapd

```
root@viejofraile:/home/user# sudo apt-get install dovecot-imapd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  dovecot-imapd
0 upgraded, 1 newly installed, 0 to remove and 267 not upgraded.
Need to get 191 kB of archives.
After this operation, 686 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 dovecot-imapd amd64 1:2.3.21+d
fsg1-2ubuntu6.1 [191 kB]
Fetched 191 kB in 1s (261 kB/s)
Selecting previously unselected package dovecot-imapd.
(Reading database ... 85%
```

Instalar Roundcube

- sudo apt-get install roundcube

Durante instalación:

- Configurar base de datos

- Seleccionar Apache2

```
root@viejofraile:/home/user# sudo apt-get install roundcube
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  dbconfig-common fonts-glyphicons-halflings javascript-common libapache2-mod-php8.3
  libjs-bootstrap libjs-bootstrap4 libjs-codemirror libjs-jquery libjs-jquery-minicolors
  libjs-jquery-ui libjs-jstimezonedetect libjs-popper.js libjs-sizzle libonig5 node-jquery
  php php-auth-sasl php-enchant php-gd php-getallheaders php-guzzlehttp-guzzle
  php-guzzlehttp-promises php-guzzlehttp-psr7 php-intl php-mail-mime php-masterminds-html5
  php-mbstring php-mysql php-net-sieve php-net-smtp php-net-socket php-pear
  php-psr-http-client php-psr-http-factory php-psr-http-message
  php-symfony-deprecation-contracts php-xml php8.3 php8.3-cli php8.3-common php8.3-enchant
  php8.3-gd php8.3-intl php8.3-mbstring php8.3-mysql php8.3-opcache php8.3-readline
  php8.3-xml roundcube-core roundcube-mysql roundcube-skin-classic roundcube-skin-larry
Suggested packages:
  dbconfig-mysql | dbconfig-pgsql | dbconfig-sqlite3 | dbconfig-no-thanks libjs-requirejs
  libjs-jquery-ui-docs php-curl php-psr-log php-laminas-httphandlerrunner roundcube-plugins
  roundcube-plugin-authres-status roundcube-plugin-contextmenu
  roundcube-plugin-html5-notifier roundcube-plugin-keyboard-shortcuts
  roundcube-plugin-listcommands roundcube-plugin-message-highlight
  roundcube-plugin-sauserprefs roundcube-plugin-thunderbird-labels php-crypt-gpg
  php-bacon-qr-code php-roundcube-rtf-html-php php-net-ldap3 php-bjeavons-zxcvbn-php
  mariadb-server | default-mysql-server
The following NEW packages will be installed:
```

10. Configurar Apache para Roundcube

- cd /etc/apache2/sites-available/
- sudo cp 000-default.conf round.conf
- sudo nano round.conf

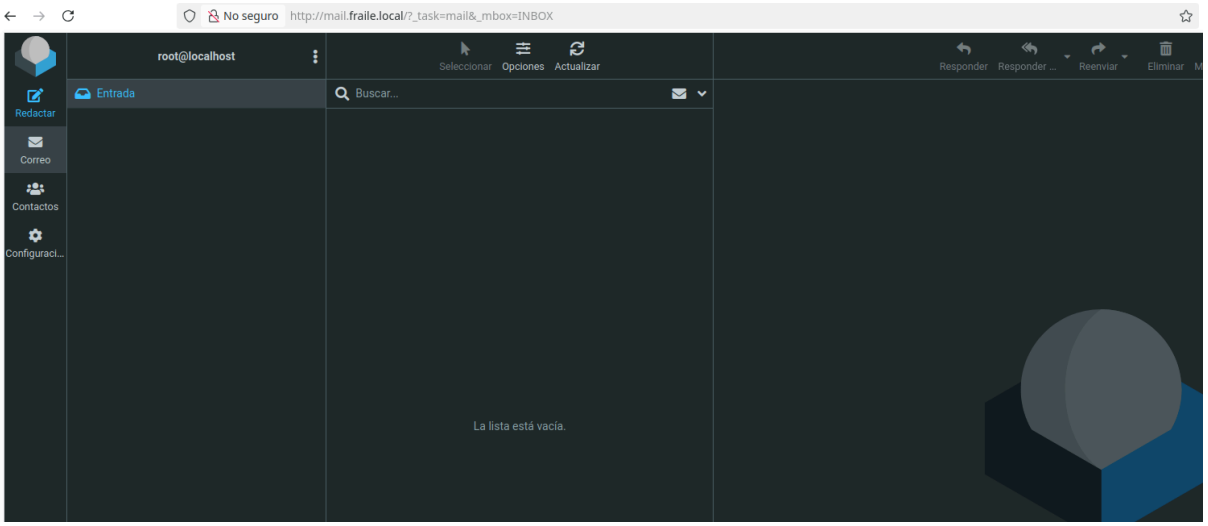
```
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        ServerName      mail.fraile.local

        ServerAdmin webmaster@localhost
        DocumentRoot /var/lib/roundcube/

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn
```

Habilitar sitio:

- sudo a2ensite round.conf
- sudo systemctl restart apache2

Acceso desde navegador:

- http://IP_DEL_SERVIDOR/



11. Añadir Registros DNS de Correo

Editar:

- sudo nano /var/cache/bind/db.tunombre.local

Agregar:

correo IN CNAME server

Reiniciar:

- sudo systemctl restart bind9

Documentacion de la pràctica

1. Comprobación y configuración:

Primero tenemos que verificar el hostname de nuestra maquina:

- hostname

Y tras esto deberemos cambiar el nombre del host usando las siguientes comandas:

- sudo hostnamectl set-hostname {dominio_principal}

```
root@abelsuarez-VirtualBox:/home/abelsuarez# hostnamectl set-hostname abelsu
arez.suarez.local
```

- sudo hostnamectl set-hostname --static {dominio_principal}

```
root@abelsuarez-VirtualBox:/home/abelsuarez# hostnamectl set-hostname --stat
ic  abelsuarez.suarez.local
```

Y confirmamos que se ha realizado el cambio nuevamente con:

- hostname

RECUERDA QUE TANTO EL HOSTNAME Y EL DOMINIO CONFIGURADO EN EL SERVIDOR DEBEN COINCIDIR.

2. Instalacion y configuracion de Postfix (Servidor SMTP)

Ahora procederemos con la instalación de postfix, para ello deberemos ejecutar la siguiente comanda:

- sudo apt-get install postfix

```
root@abelsuarez-VirtualBox:/home/abelsuarez# sudo apt-get install postfix
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer requir
ed:
  libflashrom1 libftdi1-2 libllvm13
Use 'sudo apt autoremove' to remove them.
Suggested packages:
  procmail postfix-mysql postfix-pgsql postfix-ldap postfix-pcre
  postfix-lmdb postfix-sqlite sasl2-bin | dovecot-common resolvconf
  postfix-cdb postfix-mta-sts-resolver postfix-doc
The following NEW packages will be installed:
  postfix
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
```

Saldra una interfaz donde deberemos seleccionar:

- Tipo: Internet Site
- Nombre del sistema de correo: {dominio_principal}

Una vez hecho comprobamos si el dominio se a configurado correctamente con:

- cat /etc/mailname

```
root@abelsuarez-VirtualBox:/home/abelsuarez# cat /etc/mailname
suarez.local
```

En caso de que no se haya cambiado, debes modificarlo y tras modificarlo recuerda reiniciar el servicio:

- systemctl restart postfix

```
root@abelsuarez-VirtualBox:/home/abelsuarez# systemctl restart postfix
```

3. Añadir usuarios de correo

Ahora tocará el añadir los usuarios, se crearan a partir del uso de la siguiente comanda (EN ESTE CASO LOS NOMBRES SON EJEMPLOS):

- sudo adduser **dante**
- sudo adduser **vergil**

Los usuarios creados serán capaces de enviar y recibir correo local.

```
root@abelsuarez-VirtualBox:/home/abelsuarez# adduser dante
Adding user `dante' ...
Adding new group `dante' (1001) ...
Adding new user `dante' (1001) with group `dante' ...
Creating home directory `/home/dante' ...
Copying files from `/etc/skel' ...
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
Changing the user information for dante
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
```

Contraseña: Jackpot

```
root@abelsuarez-VirtualBox:/home/abelsuarez# adduser vergil
Adding user `vergil' ...
Adding new group `vergil' (1002) ...
Adding new user `vergil' (1002) with group `vergil' ...
Creating home directory `/home/vergil' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for vergil
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
```

Contraseña: Iamthestorm

4. Instalación de Mailx (cliente de correo en consola)

Ahora procederemos con la instalacion de mailx, para ello ejecutaremos la siguiente comanda:

- sudo apt-get install bsd-mailx

```
root@abelsuarez-VirtualBox:/home/abelsuarez# apt-get install bsd-mailx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer requir
ed:
  libflashrom1 libftdi1-2 libllvm13
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  liblockfile-bin liblockfile1
The following NEW packages will be installed:
  bsd-mailx liblockfile-bin liblockfile1
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
```

5. Comprobacion de Logs del Servidor

Procederemos a la comprobacion de los logs donde verificaremos errores o envios correctos, seremos capaces de verlos al hacer un cat a la siguiente dirreccion:

- cat /var/log/mail.log

```
root@abelsuarez-VirtualBox:/home/abelsuarez# cat /var/log/mail.log
Feb 26 19:27:09 abelsuarez-VirtualBox postfix/postfix-script[6109]: starting
 the Postfix mail system
Feb 26 19:27:09 abelsuarez-VirtualBox postfix/master[6111]: daemon started -
- version 3.6.4, configuration /etc/postfix
Feb 26 19:32:39 abelsuarez postfix/postfix-script[6194]: stopping the Postfi
x mail system
Feb 26 19:32:39 abelsuarez postfix/master[6111]: terminating on signal 15
Feb 26 19:32:42 abelsuarez postfix/postfix-script[6806]: starting the Postfi
x mail system
Feb 26 19:32:42 abelsuarez postfix/master[6808]: daemon started -- version 3
.6.4, configuration /etc/postfix
```

6. Instalación de Dovecot (POP3)

Para su instalación tan solo debemos ejecutar la siguiente comanda:

- sudo apt-get install dovecot-pop3d

```
root@abelsuarez-VirtualBox:/home/abelsuarez# apt-get install dovecot-pop3d
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer requir
ed:
  libflashrom1 libftdi1-2 libllvm13
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  dovecot-core
Suggested packages:
  dovecot-gssapi dovecot-imapd dovecot-ldap dovecot-lmtpd dovecot-lucene
  dovecot-managesieved dovecot-mysql dovecot-pgsql dovecot-sieve
```

7. Configuración de POP3 en la red local

Configuración de Postfix:

Para configurar el postfix deberemos editar el archivo main.cf y agregar/modificar la siguiente linea:

- sudo nano /etc/postfix/main.cf

- mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128 [Añadir dirección de red local]
  home_mailbox = Maildir/
  mailbox_command =

```
root@abelsuarez-VirtualBox: /home/abelsuarez
  GNU nano 6.2                    /etc/postfix/main.cf

smtp_tls_CApath=/etc/ssl/certs
smtp_tls_security_level=may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache


smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defe>
myhostname = abelsuarez.suarez.local
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
myorigin = /etc/mailname
mydestination = $myhostname, suarez.local, abelsuarez.suarez.local, localho>
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128 192.168.4.0/24
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = all
home_mailbox = Maildir/
mailbox_command =
```

Y reiniciamos el servicio de postfix:

```
root@abelsuarez-VirtualBox:/home/abelsuarez# systemctl restart postfix
```

Configuracion de Dovecot:

Para su configuración deberemos editar el archivo 10-auth.conf y modificar la linea siguiente:

- sudo nano /etc/dovecot/conf.d/10-auth.conf
- disable_plaintext_auth = no

```
root@abelsuarez-VirtualBox: /home/abelsuarez
  GNU nano 6.2          /etc/dovecot/conf.d/10-auth.conf *
##
## Authentication processes
##

# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGINDISABLED capability). Note that if the remote IP
# matches the local IP (ie. you're connecting from the same computer), t
# connection is considered secure and plaintext authentication is allowe
# See also ssl=required setting.
disable_plaintext_auth = no

# Authentication cache size (e.g. 10M). 0 means it's disabled. Note that
# bsdauth and PAM require cache key to be set for caching to be used.
```

Ahora deberemos tambien de modificar el archivo 10-mail.conf y editar la siguiente linea:

- sudo nano /etc/dovecot/conf.d/10-mail.conf

- mail_location = maildir:~/Maildir

```
root@abelsuarez-VirtualBox: /home/abelsuarez
  GNU nano 6.2          /etc/dovecot/conf.d/10-mail.conf *
#   %u - username
#   %n - user part in user@domain, same as %u if there's no domain
#   %d - domain part in user@domain, empty if there's no domain
#   %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
#   mail_location = maildir:~/Maildir
#   mail_location = mbox:~/mail:INBOX=/var/mail/%u
#   mail_location = mbox:/var/mail/%d/%1n/%n:INDEX=/var/indexes/%d/%1n/%n
#
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir

# If you need to set multiple mailbox locations or want to change default
# namespace settings, you can do it by defining namespace sections.
#
```

Tras la modificación deberes reiniciar el servicio y comprobar que este se encuentra activo:

- sudo systemctl restart dovecot
- sudo systemctl status dovecot

```
root@abelsuarez-VirtualBox:/home/abelsuarez# systemctl restart dovecot
root@abelsuarez-VirtualBox:/home/abelsuarez# systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
     Loaded: loaded (/lib/systemd/system/dovecot.service; enabled; vendor p>
     Active: active (running) since Thu 2026-02-26 20:14:45 CET; 1min 1s ago
       Docs: man:dovecot(1)
             https://doc.dovecot.org/
   Main PID: 11037 (dovecot)
     Status: "v2.3.16 (7e2e900c1a) running"
      Tasks: 4 (limit: 4600)
```

8. Configuración del DNS Local

Para empezar con la configuración deberemos editar la zona directa del servidor DNS y agregar las siguientes lineas:

- sudo nano /var/cache/bind/db.{tudominio.local}

- {tudominio.local} IN MX 10 mail.{tudominio.local}.
- pop3 IN CNAME server
- smtp IN CNAME server

```
root@abelsuarez:/home/abelsuarez# cat /etc/bind/zones/db.suarez.local
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     abelsuarez.suarez.local. root.suarez.local. (
                              4         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;NS records

@       IN      NS      abelsuarez.suarez.local.

@       IN      MX      10      mail.suarez.local.
;A records

abelsuarez      IN      A       192.168.4.1
mail            IN      CNAME   abelsuarez
smtp            IN      CNAME   abelsuarez
pop3            IN      CNAME   abelsuarez
```

Y tras esto deberemos reiniciar el servicio y comprobar que las funciones funcionan usando lo siguiente:

- sudo systemctl restart bind9

```
root@abelsuarez:/home/abelsuarez# systemctl restart bind9
```

- dig -t MX {tudominio.local}

```
root@abelsuarez:/home/abelsuarez# dig -t MX suarez.local

; <<>> DiG 9.18.39-0ubuntu0.22.04.2-Ubuntu <<>> -t MX suarez.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to
 DNS
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 59483
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;suarez.local.                  IN      MX

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Feb 27 17:18:26 CET 2026
;; MSG SIZE  rcvd: 41
```

Y no nos olvidemos de limpiar la cache con:

- sudo resolvectl flush-caches

```
root@abelsuarez:/home/abelsuarez# sudo resolvectl flush-caches
```

9. Instalacion de IMAP y Webmail

Empezaremos instalando la base de datos MySQL mediante la comanda:

- sudo apt-get install mysql-server

```
root@abelsuarez:/home/abelsuarez# apt-get install mysql-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer req
uired:
  libflashrom1 libftdi1-2 libllvm13
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libaio1 libcgi-fast-perl libcgi-pm-perl libevent-core-2.1-7
  libevent-pthreads-2.1-7 libfcgi-bin libfcgi-perl libfcgi0ldbl
  libhtml-template-perl libmecab2 libprotobuf-lite23 mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0
```

Ahora procederemos a instalar el IMAP:

- sudo apt-get install dovecot-imapd

```
root@abelsuarez:/home/abelsuarez# apt-get install dovecot-imapd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer req
uired:
  libflashrom1 libftdi1-2 libllvm13
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  dovecot-imapd
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 193 kB of archives.
After this operation, 709 kB of additional disk space will be used.
```

Y también instalaremos Roundcube:

- sudo apt-get install roundcube

```
root@abelsuarez:/home/abelsuarez# apt-get install dovecot-imapd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer req
uired:
  libflashrom1 libftdi1-2 libllvm13
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  dovecot-imapd
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 193 kB of archives.
After this operation, 709 kB of additional disk space will be used.
Get:1 http://es.archive.ubuntu.com/ubuntu jammy-updates/main amd64 doveco
t-imapd amd64 1:2.3.16+dfsg1-3ubuntu2.6 [193 kB]
```

Mientras se instala deberemos seleccionar las opciones default que apareceran y seleccionaremos apache2.

En caso de que lo configures mal o te hayas equivocado ejecuta la siguiente comanda para reconfigurarlo:

- sudo dkpg-reconfigure roundcube-core

```
root@abelsuarez:/etc/apache2/sites-available# sudo dpkg-reconfigure roundcube-co
re
Determining localhost credentials from /etc/mysql/debian.cnf: succeeded.
Determining localhost credentials from /etc/mysql/debian.cnf: succeeded.
dbconfig-common: writing config to /etc/dbconfig-common/roundcube.conf
Replacing config file /etc/dbconfig-common/roundcube.conf with new version
Replacing config file /etc/roundcube/debian-db.php with new version
checking privileges on database roundcube for roundcube@localhost: user creation
 needed.
granting access to database roundcube for roundcube@localhost: success.
verifying access for roundcube@localhost: success.
dbconfig-common: dumping mysql database roundcube to /var/tmp/roundcube.roundcub
e.2026-03-05-18.46.mysql.PFT87b.
database does not exist.
dbconfig-common: dropping old mysql database roundcube.
dropping database roundcube: database does not exist.
creating database roundcube: success.
verifying database roundcube exists: success.
populating database via sql...  done.
dbconfig-common: flushing administrative password
INFO: Running /usr/share/roundcube/bin/update.sh as user 'www-data'
Replacing config file /etc/roundcube/config.inc.php with new version
```

10. Configuracion de Apache para Roundcube

Para esto deberemos dirigirnos al directorio /etc/apache2/site-available y copiar el documento 000-default.conf al documento round.conf.

- cd /etc/apache2/sites-available/
- sudo cp 000-default.conf round.conf

Tras esto modificaremos el contenido de round.conf añadiendo/modificando la información siguiente:

```
root@abelsuarez: /etc/apache2/sites-available
GNU nano 6.2                    round.conf
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and po
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header
        # match this virtual host. For the default virtual host (this file)
        # value is not decisive as it is used as a last resort host regardl
        # However, you must set it for any further virtual host explicitly.
        ServerName mail.suarez.local

        ServerAdmin webmaster@localhost
        DocumentRoot /var/lib/roundcube/public_html

        # Available loglevels: trace8, ..., trace1, debug, info, notice, wa
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example
```

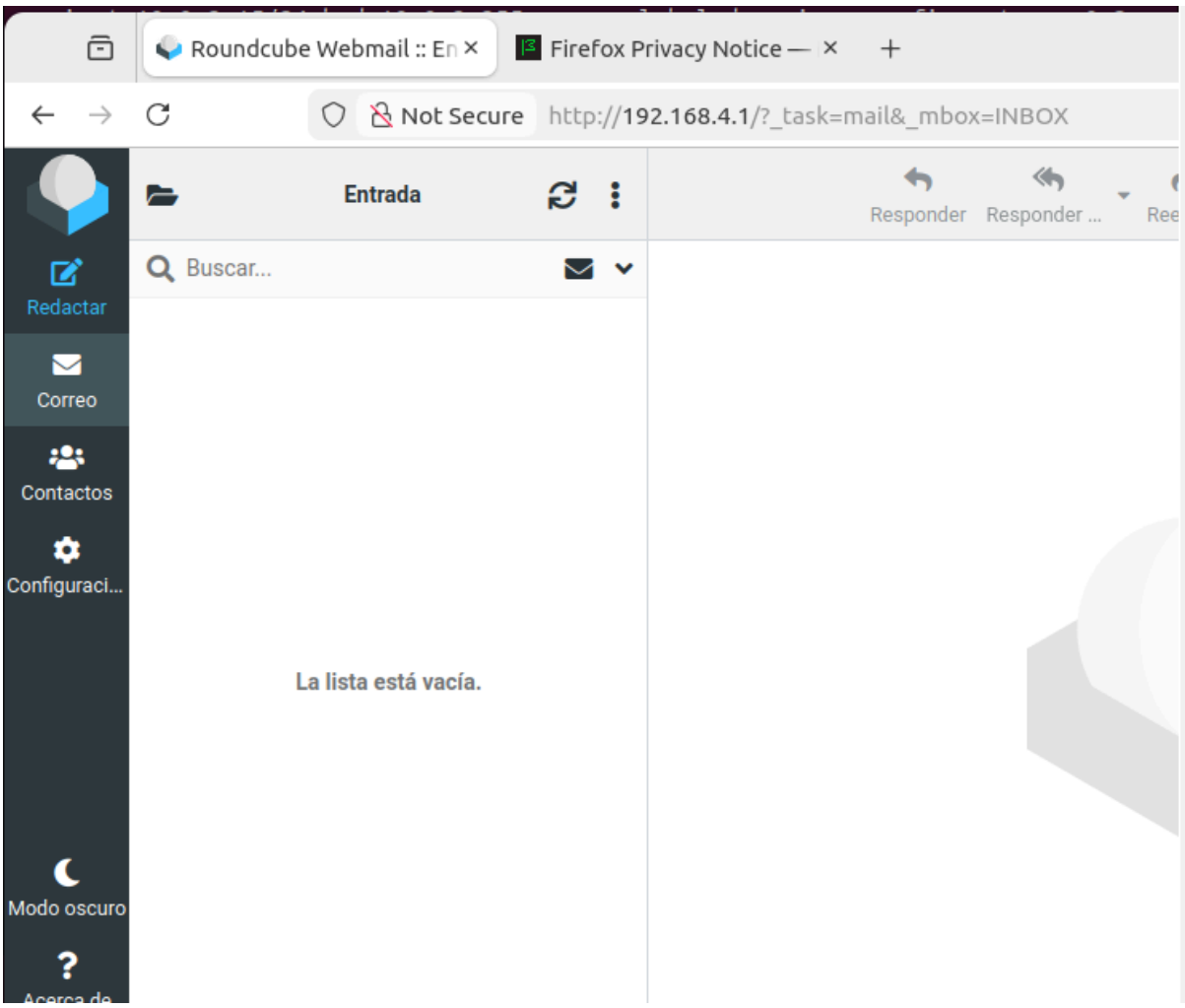
Tras esto habilitaremos el sitio usando la siguiente comanda y en caso de precaución deshabilitamos el sitio 000-default y reiniciamos el servicio:

- sudo a2ensite round.conf
- sudo a2dissite 000-default.conf
- sudo systemctl restart apache2

```
root@abelsuarez:/etc/apache2/sites-available# a2ensite round.conf
Enabling site round.
To activate the new configuration, you need to run:
  systemctl reload apache2
```

```
root@abelsuarez:/etc/apache2/sites-available# a2dissite 000-default.conf
Site 000-default disabled.
To activate the new configuration, you need to run:
  systemctl reload apache2
root@abelsuarez:/etc/apache2/sites-available# systemctl restart apache2
```

Tras haber reiniciado el servicio accederemos a la pagina web con la ip correspondiente del servidor:



11. Añadir Registros DNS de Correo

Para esto tan solo deberemos editar la zona directa y añadir la linea siguiente y tras esto reiniciaremos el servicio DNS:

- sudo nano /var/cache/bind/db.tunombre.local
- correo IN CNAME server
- sudo systemctl restart bind9

```
  GNU nano 6.2                /etc/bind/zones/db.suarez.local
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     abelsuarez.suarez.local. root.suarez.local. (
                              5         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;NS records

@       IN      NS      abelsuarez.suarez.local.

@       IN      MX      10      mail.suarez.local.
;A records

abelsuarez      IN      A       192.168.4.1
mail            IN      A       192.168.4.1
smtp            IN      CNAME   abelsuarez
pop3            IN      CNAME   abelsuarez
correo          IN      CNAME   abelsuarez
```